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Plan and context survey (Draft for deadline)

Predictive vs. Reactive Paradigms in Multi-Asset Trading

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Date: June 13, 2026

Academic Year 2025–2026

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Gantt Chart

You can find the version project and the Gantt graph by following this link, if you are authorized to access the private git:

<https://github.com/AlexandrePuisseux65/Dissertation-Project---Predictive-vs.-Reactive-Paradigms->

Literature Review

1 Context & Background

1.1 Foundations of quantitative trading

Quantitative trading, also referred to as algorithmic trading, denotes the use of automated, rule-based systems to execute buy and sell decisions across financial instruments (1). The foundations of this approach emerged in the 1970s and 1980s, when the introduction of electronic systems such as the NYSE's Designated Order Turnaround (DOT) and its successor SuperDOT enabled the automated routing of market and limit orders, significantly reducing the need for manual intervention. This technological shift marked the beginning of a broader transition away from discretionary, human-led trading towards systematic, data-driven strategies. In the early 1990s, the widespread adoption of electronic trading and Electronic Communication Networks (ECNs) shifted trading from physical floors to online platforms, laying the foundations for large-scale algorithmic trading (2).

Moreover, in 1989, the creation of the Automated Trading Desk (ATD) represented one of the earliest forms of fully automated trading, setting the tone for what would become the future of financial markets (3). High-frequency trading (HFT), an extreme form of algorithmic automation characterised by ultra-low-latency execution, emerged as a dominant force in the 2000s with the rapid development of electronic stock exchanges. By acting as primary market agents, HFT systems fundamentally transformed market microstructure and, in line with broader technological development, the speed of transactions carried out by automated systems increased considerably over the following two decades (2).

Despite variations in complexity, quantitative strategies generally follow the same core workflow. As illustrated in Figure 1, the process begins with the ingestion of various input sources, including market states, financial news, fundamental indicators, and technical signals, which are then processed by a core algorithm, function, or model. This processing stage extracts patterns and generates actionable insights, ultimately producing a trading decision as output, typically a buy, sell, long, or short signal(1).

In this sense, the logic underlying quantitative trading mirrors human decision-making: both involve processing available information to generate a prediction, which then informs an entry or exit decision (4). What distinguishes the algorithmic approach is its speed, consistency, and ability to operate without emotional bias.

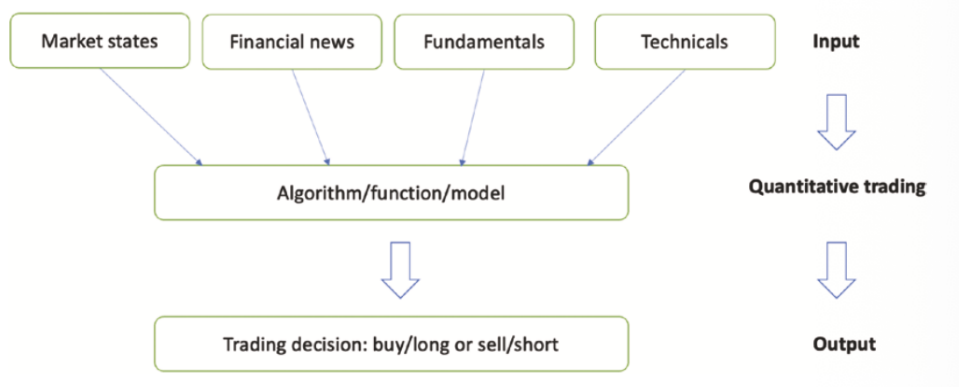


Figure 1: Illustrating the overall quantitative trading process (1).

1.2 The Rise of AI in Trading Systems

Artificial intelligence has come to occupy an increasingly prominent role in trading systems at multiple scales. Algorithms such as XGBoost, alongside deep learning architectures including Long Short-Term Memory (LSTM) networks and Convolutional Neural Networks (CNNs), have demonstrated the ability to outperform traditional quantitative models in stock price prediction and real-time decision-making (5).

Particularly useful for risk assessment and forecasting, including volatility forecasting, LSTMs and Bidirectional LSTMs excel at modeling sequential data, allowing them to map long-range dependencies in market volatility and generate highly accurate short-term risk metrics (6). Overall, advanced probabilistic models, Hidden Markov Model (HMM), and LSTMs have proven their efficacy in evaluating how assets behave under extreme, dynamic conditions.

On the other hand, in the late 2010s, Deep Reinforcement Learning (DRL) began to emerge as a promising alternative paradigm, demonstrating great adaptability in learning real-time trading policies that outperform hand-crafted strategies across stocks and futures (7). Some studies even suggest cases where DRL agents can rival professional traders on certain timeframes (8).

Current research identifies three primary directions through which DRL trading performance can be enhanced. First, enriching the state space with positional context or long-term trend modules improves both intraday and swing trading performance. Second, incorporating external information such as news sentiment or macroeconomic factors enables agents to adapt more effectively to shifting market conditions. Third, designing reward functions around risk-adjusted metrics such as the Sharpe ratio or maximum drawdown provides more robust control over capital preservation (9)(7).

However, most research still operates under simplified assumptions, with limited live trading validation and persistent challenges around data quality, robustness under regime changes, and model explainability (7). This gap between controlled backtesting performance and real-world reliability remains one of the central open problems in the field.

1.3 Challenges and Open Problems

Despite their transformative potential, AI-driven trading systems introduce a range of important technical and systemic challenges that remain only partially addressed in the current literature.

At the technical level, most models require large volumes of high-quality data to train effectively, and even then, overfitting remains a persistent problem. Models frequently perform well on historical data but fail to generalise to new market conditions, particularly in environments characterised by

high volatility or structural regime changes (10). This issue is further compounded by data latency issues, which can undermine the very speed advantage that HFT was originally designed to exploit.

Beyond individual model performance, the widespread adoption of similar AI strategies across market participants raises broader systemic concerns. Correlated trading behaviour amplifies market shocks and increases the risk of liquidity dry-ups and flash crashes (11). Furthermore, the infrastructure costs associated with deploying and maintaining AI trading systems over extended periods remains non-trivial, creating barriers to entry for smaller institutions and raising questions about the long-term economic viability of such approaches (12).

Taken together, these challenges point to a fundamental tension in the field: while AI models demonstrate strong performance under controlled backtesting conditions, their robustness, generalisability, and real-world reliability remain insufficiently validated. This observation motivates a more rigorous approach to model evaluation, which this dissertation directly addresses.

2 Predictive Approaches in Trading

2.1 LSTM and Neural Networks for Trading

Predicting financial market movements is notoriously difficult. The nature of the market and the sheer number of variables and the non-stationary nature of market dynamics made it complicated to predict efficiently. Despite this difficulty, much research actually proves that this kind of approach is possible, even if the predictive signals (the features) and future returns (the targets) are usually transparent in nature and not able to capture complex non-linear dependencies (13).

Nelson et al. (14) provided an early demonstration of LSTM applied to trading signal generation, testing the approach on Brazilian stock market data. Building on this foundation, Fischer and Krauss (13) offered a more comprehensive validation, showing that LSTM networks outperform memory-free classifiers such as random forests, deep neural networks, and logistic regression in predicting next-day cross-sectional stock returns, while generating statistically significant long-short returns.

A diverse range of predictive models is now applied to financial markets, shifting from classical statistical methods to advanced machine learning approaches. This includes ensemble tree-based models (Random Forest, XGBoost, LightGBM) and deep learning architectures tailored for time series (RNNs, LSTMs, GRUs), which demonstrate superior performance in sequential price modeling. Additionally, hybrid CNN-LSTM architectures fuse spatial feature extraction with temporal dependencies to generate richer market representations (15). More recently, image-based and trend pattern approaches have emerged, applying deep classification models directly to price chart representations to identify complex patterns beyond classical momentum signals. Finally, Deep Reinforcement Learning (DRL) agents represent a distinct paradigm, combining prediction and trading decision within a single framework that optimises directly for return or risk-adjusted metrics (an approach examined in detail in Section 3).

Among these families, this dissertation focuses specifically on LSTM networks as the predictive baseline. Beyond simple signal generation, those models have demonstrated strong potential when integrated into broader trading pipelines. Ta et al. (16) showed that combining LSTM-based price predictions with portfolio optimisation techniques such as Monte Carlo simulation and Mean-variance optimisation produced portfolios that consistently outperformed the S&P 500 benchmark over a ten-year back-test period.

2.2 From Discrete Signals to Probabilistic Outputs

Early LSTM applications in trading framed the prediction problem as a regression task, producing continuous price or return estimates evaluated primarily by error metrics such as RMSE and MSE (16). A subsequent shift reframed the problem as discrete classification, with models predicting directional outcomes such as up/down or outperform/underperform, allowing outputs to be directly translated into trading signals. Chang et al. (17) demonstrated that replacing discrete buy/sell signals with dynamic confidence thresholds improves trading signal detection, offering an early indication that probabilistic outputs carry more information than binary decisions.

Fischer and Krauss (13) exemplified this transition by training an LSTM to classify whether a stock’s next-day return would fall above or below the cross-sectional median, using a softmax output layer to produce class probabilities. Crucially, these probabilities were then used to rank assets and construct long-short portfolios, demonstrating that exploiting the full probability distribution yields higher returns and Sharpe ratios than both point-prediction models and memory-free classifiers.

More recent work has extended this probabilistic framing through hybrid architectures combining LSTM with CNN or Transformer components, further improving forecasting quality while retaining probabilistic outputs for risk-adjusted trading evaluation (18).

However, the use of probabilistic outputs primarily serves asset ranking and portfolio weighting in the existing literature. The application of confidence thresholds to filter low-conviction trades, integrate contextual market conditions, and support dynamic position sizing within a unified trading framework remains largely unexplored. This dissertation addresses this gap by transforming LSTM outputs into calibrated probabilistic classifications used directly to govern trade execution and risk management.

2.3 Limits of Predictive Models

Despite their demonstrated potential, predictive models face several well-documented limitations that constrain their practical reliability. Financial time series are inherently noisy, non-stationary, and volatile, making accurate forecasting fundamentally difficult (19). LSTM networks are particularly sensitive to hyperparameter choices, with no established guidelines for configuration, and tend to struggle with high-frequency fluctuations where short-term noise dominates the signal (20).

Beyond technical constraints, a deeper issue concerns economic validation. Li and Bastos (19) found that only 35.3% of deep learning forecasting studies analyse profitability at all, with just two implementing explicit risk management. The majority of works optimise error metrics such as RMSE or MAE while ignoring risk measures, stress tests, or performance under extreme market conditions (21).

More broadly, predictive models trained on historical data struggle to generalise across different market regimes. Authors consistently acknowledge that past performance does not guarantee future profitability, citing market noise, behavioural factors, and macroeconomic shocks as key sources of uncertainty (16). This fundamental gap between backtest accuracy and real-world robustness directly motivates the stress-testing framework developed in this dissertation.

3 Reactive Approaches in Trading

3.1 Deep Reinforcement Learning for Trading

When it emerged, this approach offered distinct advantages over traditional predictive methods by learning a trading strategy directly through market interaction, completely bypassing the need for explicit price forecasting. Instead of minimising a standard prediction error, DRL agents directly

optimise real-world financial objectives such as the Sharpe ratio, cumulative return, or maximum drawdown. This enables end-to-end decision-making that unifies feature extraction and asset allocation into a single framework, eliminating the traditional divide between signal generation and heuristic execution rules. By learning continuously from market interactions, these agents inherently adapt to changing conditions, fostering data-driven strategies that emerge organically from reward signals rather than hand-crafted rules (22)(7).

Early results confirmed this potential. Zou et al. (23) demonstrated that a cascaded LSTM-PPO system (CLSTM-PPO), tested across four major indices including the Dow Jones, S&P BSE Sensex, and FTSE100, outperformed standalone LSTM and PPO-MLP baselines by margins ranging from 5% to 52% depending on the metric and index. Similarly, Meng and Khushi (7) noted in their systematic review that DRL agents showed statistically significant profitability compared to baseline models in the majority of studies reviewed. However, they also stressed that most research remains at the proof-of-concept stage, operating under simplified assumptions such as no transaction costs, no liquidity constraints, and limited real-time validation.

3.2 Multi-Asset DRL and Portfolio Management

Extending DRL to multi-asset settings introduces additional complexity, as agents must simultaneously manage trading decisions and capital allocation across several instruments. Cheng and Sun (24) addressed this challenge through a multiagent framework combining LSTM-based price signals and CNN candlestick representations, coordinated by a portfolio manager using the TD3 algorithm. Tested across three Taiwanese stocks over 822 trading days, the system achieved consistently positive cumulative profits and win rates above 50%. However, significant drawdowns persisted, particularly for the most profitable asset, highlighting a recurring tension between return maximisation and capital preservation in multi-asset DRL systems.

Roshanpour et al. (25) further explored this tension by comparing DQN, DDPG, and PPO agents across gold, oil, and Bitcoin using Short-Time Fourier Transform (STFT) features. Their results showed that PPO combined with temporal-frequency features yielded the most robust multi-asset trader, achieving the highest risk-adjusted returns as measured by their proposed RTSS and MAI metrics. Notably, agents trained without STFT features produced high raw returns but with catastrophic drawdowns, illustrating that raw profitability alone is an insufficient measure of trading system quality.

Taken together, these studies demonstrate that while DRL agents show strong potential in multi-asset environments, each contribution evaluates its approach in isolation, under specific asset configurations and market conditions. No study directly compares DRL agents against predictive baselines within a unified benchmarking framework, nor systematically evaluates both paradigms under deliberately adverse and stress-tested conditions. This observation forms the central motivation for the comparative framework developed in this dissertation.

4 Benchmarking, Evaluation & Robustness

4.1 Evaluation Frameworks in Algorithmic Trading

Developing reliable trading systems requires not only strong predictive or reactive models, but also rigorous frameworks for evaluating their real-world performance. Pardo (26) established the methodological foundation for systematic strategy evaluation, introducing Walk-Forward Analysis as the gold standard for out-of-sample robustness testing and emphasising the importance of risk-adjusted metrics over raw profitability. This framework remains highly relevant for assessing whether

a model generalises beyond its training conditions or simply overfits historical data.

In multi-asset contexts, evaluation becomes further complicated by the need to account for diversification effects and cross-asset interactions. Peskin (27) established that multi-asset strategies require dedicated performance metrics beyond those used in single-asset evaluation, incorporating risk-adjusted returns and cross-asset correlation analysis. Despite this, the majority of algorithmic trading studies continue to evaluate models on a single asset or under a single market regime, limiting the generalisability of their conclusions.

4.2 Robustness Under Adverse Conditions

A particularly underexplored dimension of trading system evaluation is robustness under adverse market conditions. Osterrieder et al. (28) provide a comprehensive overview of stress test methodologies for AI and ML models under shifting financial conditions, identifying regime changes, volatility shocks, and distributional shifts as critical scenarios that most evaluation frameworks fail to address.

More recent work has begun to formalise adversarial testing for trading agents. Yuan et al. (29) introduced TraderBench, a framework specifically designed to evaluate AI trading agents under hostile market conditions, demonstrating that agents performing well in standard backtests often fail significantly when exposed to adversarial scenarios. Similarly, Xia et al. (30) proposed using Bayesian methods combined with adversarially generated synthetic market data to simulate extreme and rare market conditions, providing a principled approach to noise injection that goes beyond simple historical replay.

4.3 Synthesis and Research Gap

The literature reviewed across the preceding sections reveals a consistent pattern. Predictive approaches, particularly LSTM-based models, have demonstrated strong performance under controlled backtesting conditions but remain vulnerable to overfitting, regime changes, and non-stationary dynamics. Reactive approaches, particularly DRL agents, offer promising adaptability but are predominantly evaluated in isolation, under simplified market assumptions, and without systematic comparison against predictive baselines. Benchmarking frameworks, while increasingly sophisticated, have largely developed independently within each paradigm.

Tonkin et al. (31) explicitly identified this fragmentation, noting that progress in DRL trading has been hindered by the lack of cross-study comparisons and the absence of consistent evaluation conditions, and developed a framework specifically to address this within the DRL paradigm alone. Meng and Khushi (7) similarly observed that few studies provide meaningful performance comparisons between reinforcement learning and other sophisticated machine learning models. Pricope (8) further highlighted the broader benchmarking problem, noting that the diversity of experimental settings across DRL studies makes direct comparison infeasible, and explicitly called for more rigorous cross-study comparisons under consistent conditions.

While the majority of studies evaluate predictive and reactive trading systems in isolation, a small number of recent works have begun to address this limitation. Felizardo et al. (32) compared a supervised ResNet-LSTM actor against several DRL methods, demonstrating that supervised approaches can outperform RL in certain market conditions. More recently, Jomo et al. (33) proposed a unified intraday trading framework comparing regression, classification, and PPO-based RL models within a shared pipeline. However, neither study evaluates model robustness under adverse market conditions, incorporates probabilistic signal outputs, or extends the comparison to a multi-asset setting with systematic stress-testing.

This dissertation builds on these foundations by proposing a comparative framework that addresses the remaining gaps: a multi-asset evaluation under identical experimental conditions, probabilistic LSTM outputs with confidence thresholds, and systematic stress-testing through volatility shocks and artificial noise injection. Rather than claiming superiority of one paradigm over the other, the central contribution is the design and validation of a rigorous testing protocol that enables meaningful, reproducible comparison between predictive and reactive trading systems, elements that are absent from existing unified frameworks, including the most recent attempts in the literature.

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